

EEE 359

Engineering Project Management

Lecture 07

Components of Schedule Management

Precedence Diagramming Method

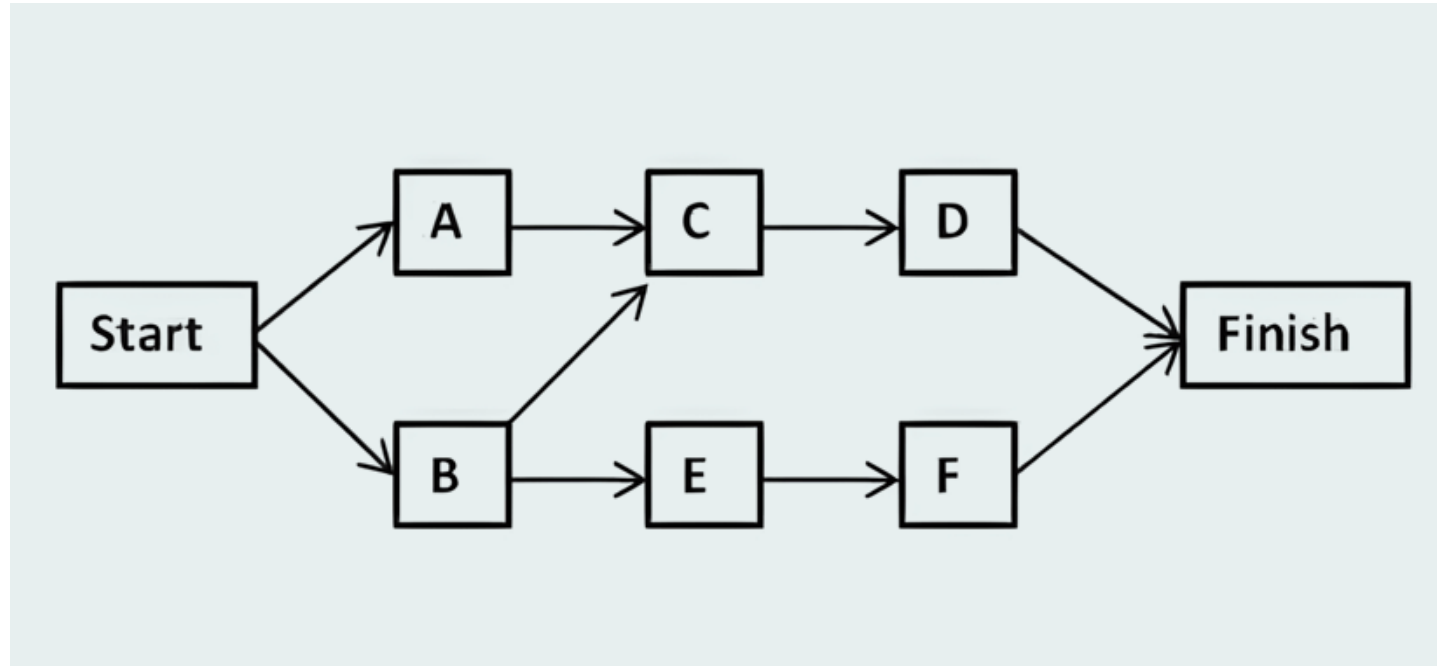
The precedence diagramming method (PDM) is a technique used for constructing a schedule model in which activities are represented by network diagrams using nodes and edges. Activities are graphically linked by one or more logical relationships to show the sequence in which the activities are to be performed.

PDM includes four types of dependencies or logical relationships.

[A predecessor activity in the logical relationship is an activity that logically comes before a dependent activity in a schedule. A successor activity is a dependent activity that logically comes after another activity in a schedule.]

- Design -> Build -> Test
- Drone design -> Hardware assembly -> Testing
-> Software Development

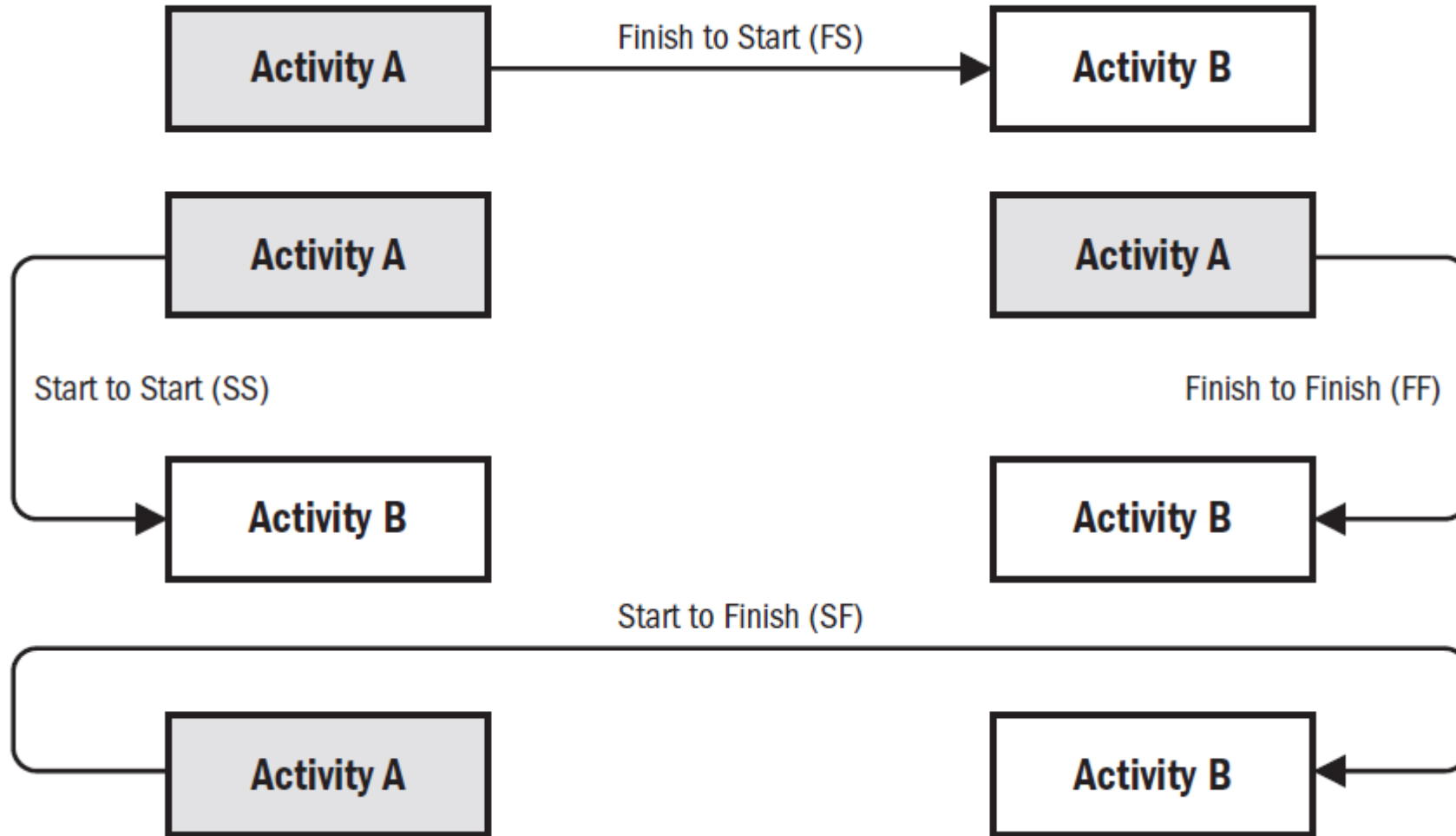
Precedence Diagramming Method



Precedence Diagramming Method

- **Finish-to-start (FS):** A logical relationship in which a successor activity cannot start until a predecessor activity has finished. For example, installing the operating system on a PC (successor) cannot start until the PC hardware is assembled (predecessor).
- **Finish-to-finish (FF).** A logical relationship in which a successor activity cannot finish until a predecessor activity has finished. For example, writing a document (predecessor) is required to finish before editing the document (successor) can finish.
- **Start-to-start (SS).** A logical relationship in which a successor activity cannot start until a predecessor activity has started. For example, level concrete (successor) cannot begin until pour foundation (predecessor) begins.
- **Start-to-finish (SF).** A logical relationship in which a successor activity cannot finish until a predecessor activity has started. For example, a new accounts payable system (successor) has to start before the old accounts payable system can be shut down (predecessor).

Precedence Diagramming Method



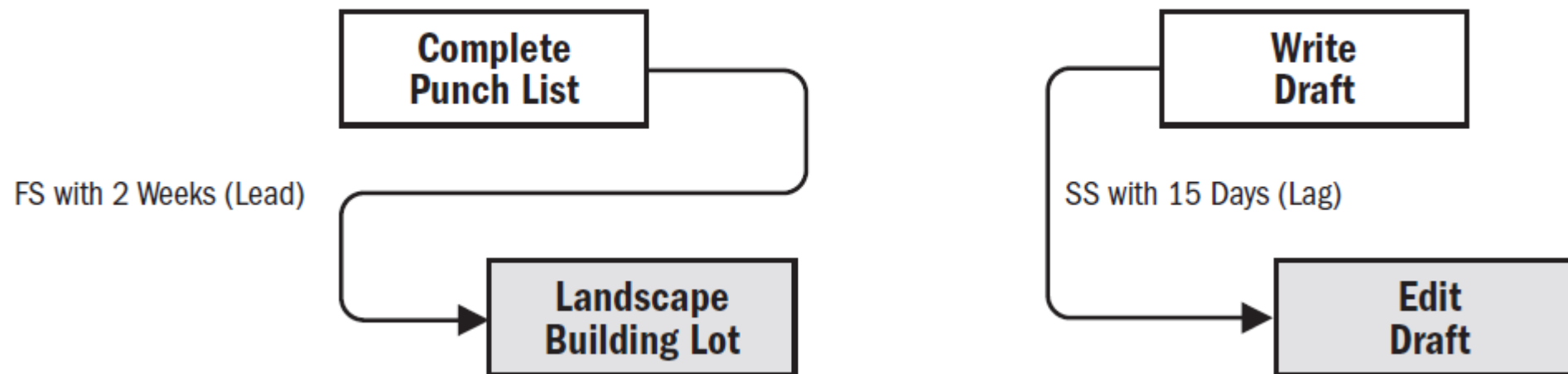
Mandatory and Discretionary Dependency

- **Mandatory dependencies** are those that are legally or contractually required or inherent in the nature of the work. Mandatory dependencies often involve physical limitations, such as on a construction project, where it is impossible to erect the superstructure until after the foundation has been built, or on an electronics project, where a prototype has to be built before it can be tested. Sometimes referred to as hard logic or hard dependencies.
- **Discretionary dependencies** are established based on best practices or preferences within a particular application area, or where a specific sequence is desired, even though other acceptable sequences may be available. Sometimes referred to as preferred logic, preferential logic, or soft logic.
- For example, best practices generally recommend completing plumbing before electrical work during construction. Although this sequence is not mandatory and both can proceed in parallel, it reduces overall project risk.
- The **project team determines** which dependencies are mandatory and which are discretionary during the process of sequencing the activities.

Leads and Lags

Lead is the amount of time a **successor activity can be advanced** with respect to a predecessor activity. For example, on a project to construct a new office building, the landscaping could be scheduled to start 2 weeks prior to the scheduled punch list completion. Lead is often represented as a negative value in scheduling software.

A lag is the amount of time a **successor activity will be delayed** with respect to a predecessor activity. For example, a technical writing team may begin editing the draft of a large document 15 days after they begin writing it.



Reserve Analysis

Reserve analysis is used to determine the duration of **contingency** and **management** reserve needed for the project.

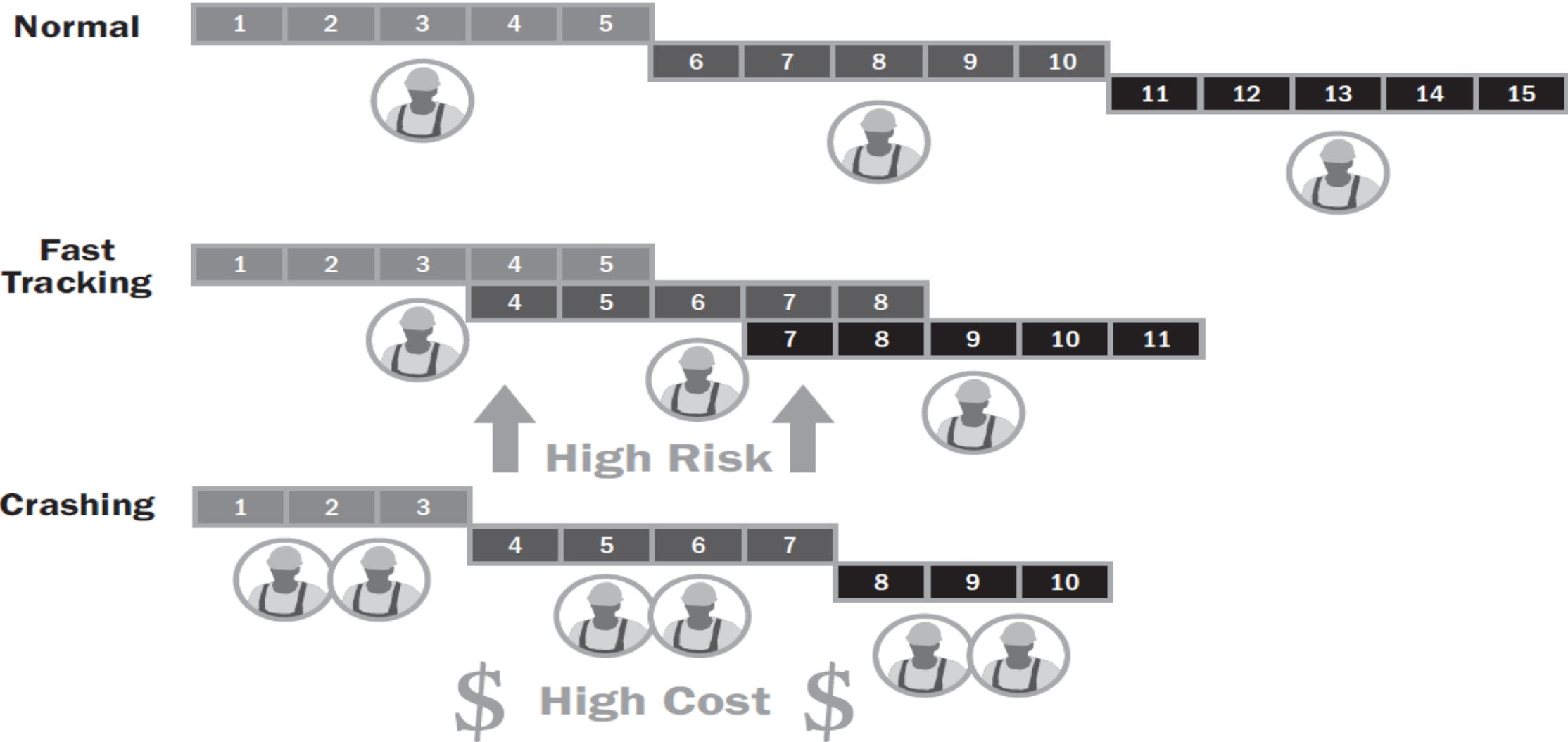
Contingency reserves are the estimated duration allocated for identified risks that are accepted. The identified risks are the **known-unknowns**, which may be estimated to account for an unknown amount of rework. The contingency reserve may be a percentage of the estimated activity duration or a fixed number of work periods. As more precise information about the project becomes available, the contingency reserve may be used, reduced, or eliminated. Contingency reserve must be included in the schedule management baseline.

Management reserves are a specified duration withheld for management control purposes and are reserved for unforeseen work that is within scope of the project. Management reserves are intended to address the **unknown-unknowns**. Management reserve is not included in the schedule management baseline, but it is part of the overall project duration requirements.

Schedule Compression

- **Crashing:** A technique used to **shorten the schedule duration** for the least incremental cost **by adding resources**. Examples of crashing include approving overtime, bringing in additional resources, or paying to expedite delivery to activities on the critical path. Crashing works only for activities on the critical path where additional resources will shorten the activity's duration. Crashing does not always produce a viable alternative and may result in increased risk and/or cost.
- **Fast tracking.** A schedule compression technique in which activities or phases normally done in **sequence** are performed in **parallel** for at least **a portion of their duration**. An example is constructing the foundation for a building before completing all of the architectural drawings. Fast tracking may result in rework and increased risk. Fast tracking only works when activities can be overlapped to shorten the project duration on the critical path. Using leads in case of schedule acceleration usually increases coordination efforts between the activities concerned and increases quality risk. Fast tracking may also increase project costs.

Schedule Compression



Thanks